

10.2.2

Design of Linear-phase FIR filters
using windows.

$$W(\omega) = \sum_{n=0}^{M-1} e^{-j\omega n}$$

Apply the geometric series formula

$$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r}, \quad \text{for } r \neq 1$$

$$= \frac{1 - e^{-j\omega M}}{1 - e^{-j\omega}} \quad \Rightarrow r = e^{-j\omega}$$

 $\Rightarrow$ Factor out $e^{-j\omega M/2}$ from Numerator & $e^{-j\omega/2}$ from denominator.

$$= \frac{e^{-j\omega M/2} (e^{j\omega M/2} - e^{-j\omega M/2})}{e^{-j\omega/2} (e^{j\omega/2} - e^{-j\omega/2})}$$

 $\Rightarrow$ use Euler's identity $\sin(x) = \frac{e^{jx} - e^{-jx}}{2j}$
 $\Rightarrow$

$$= \frac{e^{-j\omega M/2} \cdot 2j \sin(\omega M/2)}{e^{-j\omega/2} \cdot 2j \sin(\omega/2)}$$

$$W(\omega) = e^{-j\omega(M-1)/2} \cdot \frac{\sin(\omega M/2)}{\sin(\omega/2)}$$

magnitude (dB):

$$|W(\omega)| = \frac{|\sin(\omega M/2)|}{|\sin(\omega/2)|}$$

We know: $|e^{-j\omega(M-1)/2}| = 1$
 $|e^{ja}| = \sqrt{\cos^2 a + \sin^2 a} = 1$

$$\pi \leq \omega \leq 2\pi$$

Phase

$$\angle W(\omega) = \begin{cases} -\omega(\frac{M-1}{2}), & \text{when } \sin(\omega M/2) \geq 0 \\ -\omega(\frac{M-1}{2}) + \pi, & \text{when } \sin(\omega M/2) < 0 \end{cases}$$

$$W(\omega) = |W(\omega)| e^{j\theta}$$

$$\theta = -\frac{\omega(M-1)}{2}$$

Ex 10.3.4

$$H_a(s) = \frac{s + 0.1}{(s + 0.1)^2 + 16} = \frac{s + 0.1}{(s + 0.1)^2 + (4)^2}$$

$\Omega_z = 4 = \text{resonant freq.}$

we know

$$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$$

$$\Rightarrow \Omega_z = \frac{2}{T} \tan \frac{\omega_z}{2}$$

$$T \cdot 4 = 2 \tan\left(\frac{\pi}{4}\right)$$

$$T = \frac{1}{2} \cdot 1 = \frac{1}{2}$$

$$\frac{\pi}{2} \cdot \frac{1}{2} = \frac{\pi}{4}$$

$$\tan\left(\frac{\pi}{4}\right) = 1$$

we know $\rightarrow$

$$\Omega = \frac{2}{T} \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right)$$

$$= 4 \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right) \text{ put this in eq. 1}$$

$$H(z) = \frac{4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.1}{\left(\frac{4 \cdot \frac{1-z^{-1}}{1+z^{-1}} + 0.1 \right)^2 + 16}$$

numerator:

$$= \frac{4(1-z^{-1}) + 0.1(1+z^{-1})}{1+z^{-1}} = \frac{4 - 4z^{-1} + 0.1 + 0.1z^{-1}}{1+z^{-1}}$$

$$= \frac{4.1 - 3.9z^{-1}}{1+z^{-1}} \quad (1)$$

Den:

$$\left(\frac{4.1 - 3.9z^{-1}}{1+z^{-1}} \right)^2 + 16 = \frac{(4.1 - 3.9z^{-1})^2 + 16(1+z^{-1})^2}{(1+z^{-1})^2}$$

$$= \frac{16.81 + 32.21z^{-2} - 31.98z^{-1} + 16 + 32z^{-1} + 16z^{-2}}{(1+z^{-1})^2}$$

$$= \frac{32.81 + 0.02z^{-1} + 32.21z^{-2}}{(1+z^{-1})^2}$$

Combine num & den:

$$= \frac{4.1 - 3.9z^{-1}}{1+z^{-1}} \cdot \frac{(1+z^{-1})^2}{32.81 + 0.02z^{-1} + 32.21z^{-2}}$$

$$= \frac{4.1 + 0.2z^{-1} - 3.9z^{-2}}{32.81 + 0.02z^{-1} + 32.21z^{-2}}$$

÷ num & den by 32.81 to make constant term in den = 1.

$$H(z) = \frac{0.125 + 0.0061z^{-1} - 0.1189z^{-2}}{1 + 0.0006z^{-1} + 0.975z^{-2}}$$

extremely small hence,

$$H(z) = \frac{0.125 + 0.0061z^{-1} - 0.1189z^{-2}}{1 + 0.975z^{-2}}$$

ply num & den by z^2 to remove negative powers:

$$H(z) = \frac{0.125z^2 + 0.0061z - 0.1189}{z^2 + 0.975}$$

⇒ the filter has poles at $P_{1,2} = 0.987 \pm j0.177$

⇒ the filter has zeros at $Z_{1,2} = -1.095$